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**PHYSICS**

**0625/41**

Paper 4 Extended Theory

**May/June 2019**

MARK SCHEME

Maximum Mark: 80

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**Published**

This mark scheme is published as an aid to teachers and candidates, to indicate the requirements of the examination. It shows the basis on which Examiners were instructed to award marks. It does not indicate the details of the discussions that took place at an Examiners' meeting before marking began, which would have considered the acceptability of alternative answers.

Mark schemes should be read in conjunction with the question paper and the Principal Examiner Report for Teachers.

Cambridge International will not enter into discussions about these mark schemes.

Cambridge International is publishing the mark schemes for the May/June 2019 series for most Cambridge IGCSE™, Cambridge International A and AS Level and Cambridge Pre-U components, and some Cambridge O Level components.

**PUBLISHED****Generic Marking Principles**

These general marking principles must be applied by all examiners when marking candidate answers. They should be applied alongside the specific content of the mark scheme or generic level descriptors for a question. Each question paper and mark scheme will also comply with these marking principles.

**GENERIC MARKING PRINCIPLE 1:**

Marks must be awarded in line with:

- the specific content of the mark scheme or the generic level descriptors for the question
- the specific skills defined in the mark scheme or in the generic level descriptors for the question
- the standard of response required by a candidate as exemplified by the standardisation scripts.

**GENERIC MARKING PRINCIPLE 2:**

Marks awarded are always **whole marks** (not half marks, or other fractions).

**GENERIC MARKING PRINCIPLE 3:**

Marks must be awarded **positively**:

- marks are awarded for correct/valid answers, as defined in the mark scheme. However, credit is given for valid answers which go beyond the scope of the syllabus and mark scheme, referring to your Team Leader as appropriate
- marks are awarded when candidates clearly demonstrate what they know and can do
- marks are not deducted for errors
- marks are not deducted for omissions
- answers should only be judged on the quality of spelling, punctuation and grammar when these features are specifically assessed by the question as indicated by the mark scheme. The meaning, however, should be unambiguous.

**GENERIC MARKING PRINCIPLE 4:**

Rules must be applied consistently e.g. in situations where candidates have not followed instructions or in the application of generic level descriptors.

**GENERIC MARKING PRINCIPLE 5:**

Marks should be awarded using the full range of marks defined in the mark scheme for the question (however; the use of the full mark range may be limited according to the quality of the candidate responses seen).

**GENERIC MARKING PRINCIPLE 6:**

Marks awarded are based solely on the requirements as defined in the mark scheme. Marks should not be awarded with grade thresholds or grade descriptors in mind.

| Question | Answer                                                                                                                                      | Marks     |
|----------|---------------------------------------------------------------------------------------------------------------------------------------------|-----------|
| 1(a)     | change of velocity per unit time <b>OR</b> $\frac{v-u}{t}$                                                                                  | <b>B1</b> |
| 1(b)     | line starts at origin <b>and</b> is asymptotic to x-axis                                                                                    | <b>B1</b> |
|          | increasing gradient initially <b>and</b> no decrease                                                                                        | <b>B1</b> |
|          | constant <b>and</b> clearly positive gradient finally                                                                                       | <b>B1</b> |
| 1(c)(i)  | no external forces <b>OR</b> isolated system                                                                                                | <b>B1</b> |
|          | sum of momenta / (total) momentum remains constant                                                                                          | <b>B1</b> |
| 1(c)(ii) | rocket <u>gains</u> (upward) momentum                                                                                                       | <b>B1</b> |
|          | (ejected) gas <u>gains</u> equal (quantity of) momentum in opposite direction<br><b>OR</b> momentum of gas <u>decreases</u> by equal amount | <b>B1</b> |

| Question | Answer                                                                       | Marks     |
|----------|------------------------------------------------------------------------------|-----------|
| 2(a)     | $(W =) mg$ <b>OR</b> $3.4 \times 10^3 \times 10$                             | <b>C1</b> |
|          | $3.4 \times 10^4 \text{ N}$                                                  | <b>A1</b> |
| 2(b)(i)  | moment = $Fx$ in any form <b>OR</b> (moment) = $Fx$ <b>OR</b> 0.50 (seen)    | <b>C1</b> |
|          | $3.4 \times 10^4 \times (1.8 - 1.3)$ <b>OR</b> $3.4 \times 10^4 \times 0.50$ | <b>C1</b> |
|          | $1.7 \times 10^4 \text{ N m}$                                                | <b>A1</b> |
| 2(b)(ii) | 1. (the point) where (all) the mass can be considered to be concentrated     | <b>B1</b> |
|          | 2. $1.7 \times 10^4 / (1.3 + 0.70)$ <b>OR</b> $1.7 \times 10^4 / (2.0)$      | <b>C1</b> |
|          | $8.5 \times 10^3 \text{ N}$                                                  | <b>A1</b> |
| 2(c)     | (moment / it) increases                                                      | <b>B1</b> |
|          | perpendicular distance (between P and line of action of) $W$ increases       | <b>B1</b> |

| Question | Answer                                                                                                                                                                                                 | Marks     |
|----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|
| 3(a)     | (air) molecules / they move / collide                                                                                                                                                                  | <b>B1</b> |
|          | (air) molecules / they collide with cube / (upper) surface (of cube) / wall                                                                                                                            | <b>B1</b> |
|          | impulse exerted (on surface) <b>OR</b> momentum change (of molecules)                                                                                                                                  | <b>B1</b> |
| 3(b)(i)  | $p = h\rho g$ in any form <b>OR</b> $(p =) h\rho g$ <b>OR</b> $0.028 \times 1500 \times 10$                                                                                                            | <b>C1</b> |
|          | 420 Pa                                                                                                                                                                                                 | <b>A1</b> |
| 3(b)(ii) | $F = pA$ in any form words, symbols or numbers <b>OR</b> $(F =) pA$<br><b>OR</b> $420 \times 4.0^2$ <b>OR</b> $420 \times 0.040^2$ <b>OR</b> $420 \times 16$ <b>OR</b> $420 \times 1.6 \times 10^{-3}$ | <b>C1</b> |
|          | 0.67 N                                                                                                                                                                                                 | <b>A1</b> |
| 3(c)(i)  | $W = Fd$ in any form words, symbols or numbers <b>OR</b> $(W =) Fd$<br><b>OR</b> $0.67 \times 0.034$                                                                                                   | <b>C1</b> |
|          | 0.023                                                                                                                                                                                                  | <b>A1</b> |
| 3(c)(ii) | lifting liquid as well <b>OR</b> friction between liquid and container / pipe                                                                                                                          | <b>B1</b> |

| Question | Answer                                                                                                                                         | Marks     |
|----------|------------------------------------------------------------------------------------------------------------------------------------------------|-----------|
| 4(a)(i)  | $E = mc (\Delta)T$ in any form words, symbols or numbers <b>OR</b> $(E =) mc (\Delta)T$<br><b>OR</b> $0.23 \times 0.72 \times 550$             | <b>C1</b> |
|          | 91 J                                                                                                                                           | <b>A1</b> |
| 4(a)(ii) | 1. $t = E / P$ in any form words, symbols or numbers <b>OR</b> $(t =) E / P$ <b>or</b> $91 / 2.4$                                              | <b>C1</b> |
|          | 38 s                                                                                                                                           | <b>A1</b> |
|          | 2. (thermal) energy is used to increase the temperature of / lost to cylinder / piston / heater / surroundings                                 | <b>B1</b> |
| 4(b)(i)  | it / piston moves to the <u>right</u> / <u>away from heater</u> <b>OR</b> <u>accelerates</u> (to right)                                        | <b>M1</b> |
|          | pressure (of gas) greater / pressure greater (on left) / <u>resultant</u> force to right                                                       | <b>A1</b> |
| 4(b)(ii) | $V_2 = p_1 V_1 / p_2$ in any form <b>OR</b> $(V_2 =) p_1 V_1 / p_2$<br><b>OR</b> $2.9 \times 10^5 \times 1.9 \times 10^{-4} / 1.0 \times 10^5$ | <b>C1</b> |
|          | $5.5 \times 10^{-4} \text{ m}^3$                                                                                                               | <b>A1</b> |

| Question | Answer                                                                                                                                                                                                               | Marks     |
|----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|
| 5(a)(i)  | any <b>two</b> from:<br><br>occurs throughout the liquid <b>OR</b> bubbles formed<br>occurs at one temperature / boiling point<br>does not produce cooling <b>OR</b> unaffected by draught / surface area / humidity | <b>B2</b> |
| 5(a)(ii) | (more) energetic molecules escape (from the liquid) <b>OR</b> molecules gain energy and escape <b>OR</b> molecules overcome intermolecular forces / break bonds                                                      | <b>B1</b> |
|          | average speed decreases <b>OR</b> molecules with less (kinetic) energy left behind                                                                                                                                   | <b>B1</b> |
|          | temperature of liquid decreases                                                                                                                                                                                      | <b>B1</b> |
|          | (thermal) energy conducted / gained from skin / body <b>OR</b> (thermal) energy lost by skin / body                                                                                                                  | <b>B1</b> |
| 5(b)     | molecules touching <b>OR</b> no space between molecules                                                                                                                                                              | <b>B1</b> |
|          | large (repulsive / intermolecular) forces (when moved closer)                                                                                                                                                        | <b>B1</b> |



| Question | Answer                                                                                                                                                                                                                    | Marks     |
|----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|
| 6(a)     | idea of one side of wavefront enters / hits solid first <b>OR</b> wavefront does not all hit the solid all at once;                                                                                                       | <b>B1</b> |
|          | idea of this side slowed down first <b>OR</b> this side delayed relative to other side                                                                                                                                    | <b>B1</b> |
|          | angle of wave(front) changes <b>OR</b> different parts of wavefront delayed by different amounts                                                                                                                          | <b>B1</b> |
| 6(b)(i)  | $n = \frac{\sin i}{\sin r}$ in any form <b>OR</b> $n_1 \sin \theta_1 = n_2 \sin \theta_2$ <b>OR</b> $1.3 = \frac{\sin 67^\circ}{\sin r}$<br><b>OR</b> $(r = ) \sin^{-1}(\sin 67^\circ / 1.3)$ <b>OR</b> $\sin^{-1}(0.71)$ | <b>C1</b> |
|          | 45°                                                                                                                                                                                                                       | <b>A1</b> |
| 6(b)(ii) | $v_{ts} = c / n$ in any form <b>OR</b> $(v_{ts} =) c / n$ <b>OR</b> $3.0 \times 10^8 / 1.3$                                                                                                                               | <b>C1</b> |
|          | $2.3 \times 10^8$ <b>OR</b> $3.0 \times 10^8 / 1.3$                                                                                                                                                                       | <b>C1</b> |
|          | $\lambda = v / f$ in any form <b>OR</b> $(\lambda =) v / f$ <b>OR</b> $2.3 \times 10^8 / 5.7 \times 10^{14}$<br><b>OR</b> $3.0 \times 10^8 / (1.3 \times 5.7 \times 10^{14})$                                             | <b>C1</b> |
|          | $4.0 \times 10^{-7} \text{ m}$                                                                                                                                                                                            | <b>A1</b> |
|          | <b>OR</b> (alternative approach)                                                                                                                                                                                          |           |
|          | $\lambda = v / f$ in any form <b>OR</b> $(\lambda =) v / f$ <b>OR</b> $3.0 \times 10^8 / 5.7 \times 10^{14}$                                                                                                              | <b>C1</b> |
|          | $5.3 \times 10^{-7}$ <b>OR</b> $3.0 \times 10^8 / 5.7 \times 10^{14}$                                                                                                                                                     | <b>C1</b> |
|          | $\lambda_g = \lambda_a / n$ in any form <b>OR</b> $(\lambda_g =) \lambda_a / n$ <b>OR</b> $5.3 \times 10^{-7} / 1.3$<br><b>OR</b> $3.0 \times 10^8 / (1.3 \times 5.7 \times 10^{14})$                                     | <b>C1</b> |
|          | $4.0 \times 10^{-7} \text{ m}$                                                                                                                                                                                            | <b>A1</b> |

| Question | Answer                                                                                                                                                                       | Marks     |
|----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|
| 7(a)     | thermistor <b>c.a.o.</b>                                                                                                                                                     | <b>B1</b> |
| 7(b)(i)  | $V_X = V_{30}$                                                                                                                                                               | <b>B1</b> |
| 7(b)(ii) | $V_X = E - V_{20}$ in any form                                                                                                                                               | <b>B1</b> |
| 7(c)(i)  | $1/R_1 + 1/R_2 = 1/R_{\text{tot}}$ <b>OR</b> $(R_{\text{tot}} =) R_1 R_2 / (R_1 + R_2)$ <b>OR</b> $1/15 + 1/30 = 1/R_{\text{tot}}$<br><b>OR</b> $(15 \times 30) / (15 + 30)$ | <b>C1</b> |
|          | 10 ( $\Omega$ ) <b>OR</b> 10 + 20                                                                                                                                            | <b>C1</b> |
|          | 30 $\Omega$                                                                                                                                                                  | <b>A1</b> |
| 7(c)(ii) | $I = V / R$ in any form <b>OR</b> $(I =) V / R$ <b>OR</b> 6.0 / 30                                                                                                           | <b>C1</b> |
|          | 0.20 A                                                                                                                                                                       | <b>A1</b> |
| 7(d)     | resistance <u>of</u> <u>X</u> decreases                                                                                                                                      | <b>B1</b> |
|          | ammeter reading / it increases <b>and</b> (total) resistance (of circuit) decreases / more voltage across 20 $\Omega$ resistor                                               | <b>B1</b> |

| Question | Answer                                                                                                       |           |                                                                        | Marks                  |
|----------|--------------------------------------------------------------------------------------------------------------|-----------|------------------------------------------------------------------------|------------------------|
| 8(a)(i)  | <u>magnetic</u> field mentioned                                                                              |           |                                                                        | <b>B1</b>              |
|          | coil / wire cuts (magnetic) field <b>OR</b> changing (magnetic) field (through coil)                         |           |                                                                        | <b>B1</b>              |
|          | e.m.f. / voltage <u>induced</u> <b>OR</b> produced by electromagnetic <u>induction</u>                       |           |                                                                        | <b>B1</b>              |
| 8(a)(ii) | (plane of coil) horizontal <b>OR</b> in position shown in diagram<br>coil cutting magnetic field the fastest |           |                                                                        | <b>B1</b><br><b>B1</b> |
| 8(b)     | current in coil                                                                                              | <b>OR</b> | energy supplied to / lost from lamp                                    | <b>B1</b>              |
|          | current in (magnetic) field experiences a force                                                              |           | student must do more work / supply more energy /<br>more energy needed | <b>B1</b>              |
|          | opposes the change causing it                                                                                |           | greater force to do more work                                          | <b>B1</b>              |

| Question  | Answer                                                                                                                  | Marks     |
|-----------|-------------------------------------------------------------------------------------------------------------------------|-----------|
| 9(a)(i)   | <b>mark both explanation and deduction together</b>                                                                     |           |
|           | nucleus is very small                                                                                                   | <b>B1</b> |
|           | very few $\alpha$ -particles hit or pass near to a nucleus                                                              | <b>B1</b> |
| 9(a)(ii)  | <b>mark both explanation and deduction together</b>                                                                     |           |
|           | <u>nucleus</u> is charged                                                                                               | <b>B1</b> |
|           | (charged) $\alpha$ -particles experience a force                                                                        | <b>B1</b> |
| 9(a)(iii) | <b>mark both explanation and deduction together</b>                                                                     |           |
|           | centre / (small) part of atom <b>OR</b> nucleus includes most of the mass of the atom / is (very) dense                 | <b>B1</b> |
|           | ( $\alpha$ -particles move and) nucleus stays still                                                                     | <b>B1</b> |
| 9(b)      | any <b>two</b> from:<br><br>opposite direction<br>(much) smaller deflection<br>undergo deflections of similar magnitude | <b>B2</b> |